ADDITIONAL INFORMATION

Course: DLMDSAS01 – Advanced Statistics

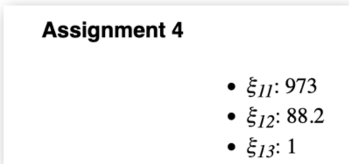
1. STRUCTURE OF THE WORKBOOK:

The structure of each of the workbook should reflect a logical progression. The text should be a coherent and cohesive whole and follow a linear structure. There should be a logical integration of each of the points across the various sections, with clear transitions between them. Referring to a fact in a previous paragraph, a theorem in a book, or a calculation in another place is necessary to perform a valid reasoning step.

Each of the assignments can be solved in isolation but should be presented in a single section. The usage of any lemma, formula, theorem or data should be made with a reference. Internal references (such as: see equation 3.17) are welcome: Every theorem, definition or method that you apply should be mentioned as a reference, including a paragraph, theorem, or definition number.

The solution to each assignment is expected to be created by yourself. For this reason everyone has a single set of numbers of the assignment which is generated by an online service: the parameter generator.

The parameter generator provides numbers that are behind each of the $\xi_1, \ldots, \xi_n$ variables within the assignment. For example, in the exercise 4 if having the parameters obtained in Figure 1, you are expected to read the first sentence as:



> Over a long period of time, the production of 1000 high-quality hammers in a factory seems to have reached a weight with an average of $\mathbf{973}$ $(\mathbf{in}\ \boldsymbol{g})$ and standard deviation of $\mathbf{88.2}$ $(\mathbf{in}\ \boldsymbol{g})$.

1 Photo: P Libbrecht

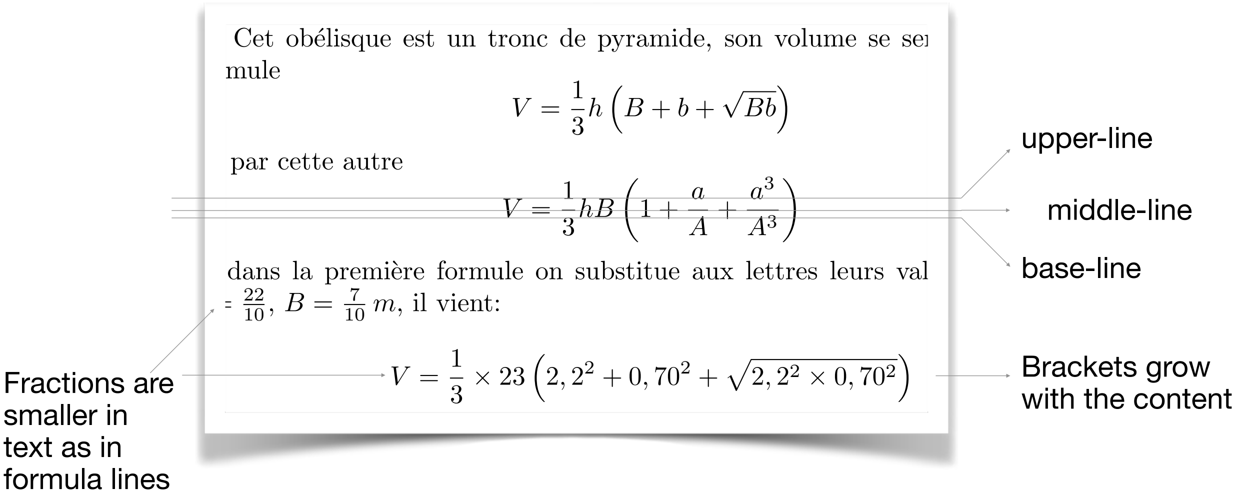
Please include in your assignment the generated signature string as well as the parameters provided by the online service. A submission without dynamically generated parameters and a signature will not be considered valid.

2. MATHEMATICAL ARGUMENTATION

The exercises in the workbook are all mathematical in nature: You are expected to draw conclusions from given information following a completely documented argumentation path. If you need a refresher on mathematical proofs, a short reminder Hutchings (2003) is available to you.

Hutchings, M. (2003): *Introduction to mathematical arguments.* Available from https://math.berkeley.edu/~hutching/teach/proofs.pdf (last checked 2020-06-25).

Mathematical argumentation also uses mathematical notations and this is an important part of the vocabulary. You find in the figure below with mathematical notations with a few important concepts of the notations: exponents, fractions, brackets, roots. Some other notation vocabulary elements such as large-operators (such as integrals), tables (such as matrices), decoration on variables (e.g. an arrow for a vector or a hat for a maximum), or commutative diagrams.


Cet obélisque est un tronc de pyramide, son volume se se
mule

$$V = \frac{1}{3}h\left(B + b + \sqrt{Bb}\right)$$

par cette autre

$$V = \frac{1}{3}hB\left(1 + \frac{a}{A} + \frac{a^3}{A^3}\right)$$

dans la première formule on substitue aux lettres leurs val
$= \frac{22}{10}$, $B = \frac{7}{10}\,m$, il vient:

$$V = \frac{1}{3} \times 23\left(2,2^2 + 0,70^2 + \sqrt{2,2^2 \times 0,70^2}\right)$$

(retyped from From André, M. Ph. (1895), Nouveau cours de géométrie, Librairie André Guedon. https://gallica.bnf.fr/ark:/12148/bpt6k64643463/)

From this little example one sees that mathematical notations uses vertical and horizontal constructs intensively to express its meanings. You are required to use 2d-mathematical notations in the workbook. Please use regular mathematical notations or declare the notations you use in advance.

3. FORMATTING

Templates in Word and LaTeX are made available in myCampus with a complete set of styles for each of the sections. Please make use of these templates as they provide a logical structure for the content. Note that LaTeX templates tends to produce slightly higher quality PDFs.

Also, please make sure that mathematical formulas are typed using the mathematical tools (Insert > Equation in Word or using $ or $$ in LaTeX). Any calculation should be detailed, including the steps, in the assignment. If using computer-based tools to perform calculations (e.g. R to create a plot or WolframAlpha to compute a derivative), please include the source code or URL used so that your calculations can be reproduced and understood. Make sure that this piece of code can be copied from the PDF (e.g. do not include it as a PNG picture) as the PDF is the only submission content.

4. CREATING GRAPHS

In order to draw the graphs, you may use any generic plotting tool or any drawing tool based on numbers calculated by hand or by a software program. An introduction to using Plotly (a popular program for creating web-based graphics based on data provided in arrays) is available on myCampus.